Notes: Peress and Schmidt (JF, 2020)

Glued to the TV: Distracted Noise Traders and Stock Market Liquidity

Contents

1 Big picture

- **Question.** What happens to **market liquidity, volatility, and price efficiency** when **noise traders** become inattentive and temporarily stop trading?
- **Setting.** The paper studies **U.S. equity markets** using episodes of **sensational noneconomic news**—for example, the O.J. Simpson verdict, terrorist attacks, celebrity deaths, natural disasters, and major accidents—that distract investors from the stock market.
- **Core challenge.** Most attention shocks are confounded with **stock-relevant news**. If investors pay less attention because of firm news, macro news, or price moves, we cannot tell whether any market effect comes from attention itself or from fundamentals. The paper therefore needs shocks that are:
  1. **exogenous to the stock market**,
  2. **attention-diverting** rather than attention-attracting, and
  3. targeted mostly at **noise traders**, not informed traders.
- **Main methodological contribution.** The paper uses **high-news-pressure days** from U.S. television news and then filters out days containing economic keywords. This yields a set of **551 distraction events** from 1968 to 2014 that plausibly capture exogenous shocks to investor attention.
- **Main economic contribution.** The paper shows that the attention of **noise traders** matters for **market liquidity**. When noise traders are distracted and trade less, **adverse selection becomes worse** for market makers, so liquidity falls.
- **Why it matters.** The paper clarifies an important ambiguity in market microstructure. A decline in noise trading can either improve liquidity through lower inventory risk or worsen liquidity through stronger adverse selection. The answer depends on whether the noise-trading shock is **persistent** or **short-lived**.
- **Main empirical question.** On days when sensational noneconomic news distracts households and institutions, what happens to:
  - trading activity,
  - liquidity,
  - volatility, and
  - price reversals?
- **Main empirical results.**
  - On distraction days, **household trading volume falls by about 6.5%** and **institutional trading volume falls by about 4%**.
  - Among retail investors, the main effect is on the **extensive margin**: the number of investors who trade falls by about **5%**.
  - Small trades fall more than large trades, consistent with distraction hitting **retail / noise trading** more than institutional informed trading.

- For stocks with high retail ownership—small stocks, low-price stocks, and low-institutional-ownership stocks—trading activity declines by roughly **3%**, liquidity worsens by about **1%–6%**, volatility falls by about **3%**, and prices reverse less.
- Cross-sectional event tests show that events with larger drops in trading also exhibit larger drops in volatility and larger increases in illiquidity.
- Distraction effects become stronger after the rise of the **Internet**, and after 2007 they become especially strong in stocks with high levels of **algorithmic trading**.

- **Economic takeaway.** Noise traders do not simply create useless volume. Their presence **masks informed trading** and thereby helps market makers bear adverse-selection risk. When noise traders step away, markets become **quieter but less liquid**.
- **Connection to the literature.** The paper adds to work on limited attention by shifting focus away from informed traders and market makers toward **noise traders**. It also reconciles competing findings in the liquidity literature by showing that **transitory** drops in noise trading worsen liquidity, whereas **persistent** drops can improve it.

2 Data and measurement (key objects)

2.1 Distracting events and the construction of the shock

- **Core event variable.** The paper starts from **news pressure**, introduced by Eisensee and Strömberg. News pressure is the median number of minutes that U.S. evening news broadcasts devote to their first three news segments.
- **Initial screening rule.** For each year, the authors select the **top 10%** of business days in terms of news pressure. This yields **1,084 candidate days**.
- **Economic-news filter.** The authors then discard days whose top-news headlines contain economic keywords such as *banking, bankruptcy, federal reserve, interest rate, stock market, treasury, recession, inflation*, and so on.
- **Final sample.** After filtering, the final sample contains **551 distraction days** from **1968 to 2014**.
- **Examples.** The event list includes the O.J. Simpson verdict, the Challenger explosion, the Oklahoma City bombing, Princess Diana's funeral, the Virginia Tech massacre, Hurricane Katrina, and the Haiti earthquake.
- **Why this variable is useful.** The paper wants stories that draw mass attention but are plausibly orthogonal to future cash flows and discount rates. These TV-news shocks are designed to satisfy exactly that requirement.

2.2 Evidence that the events really attract attention

- **TV viewership data.** Using Nielsen data for 1991–2014, the paper shows that viewership rises sharply on distraction days.
- **Main fact.**

  - CNN viewership rises by about **33%**.
  - ABC/CBS/NBC evening-news viewership rises by about **3%**.
- **Interpretation.** These days truly absorb household attention; investors are literally more likely to be **"glued to the TV."**

2.3 Trader-level data

- **Retail brokerage data.**
  - About **1.9 million** common-stock trades.
  - Roughly **78,000** households.
  - Period: **January 1991 to November 1996**.
  - Number of distraction events in this window: **66**.
- **Institutional trade data (ANcerno / Abel Noser).**
  - Institutional transactions by **835 asset managers**.
  - Period: **1999 to mid-2011**.
  - Number of distraction events: **99**.
- **Aggregated transaction data (ISSM/TAQ).**
  - Covers all NYSE/AMEX/Nasdaq transactions.
  - Used mainly for **small versus large trade** comparisons.
  - Period: **1991–2000**.
  - Number of distraction events: **105**.

2.4 Stock-market data

- **CRSP daily data.**
  - Main market-wide and stock-level daily outcomes.
  - Period: **1968–2014**.
  - Number of distraction events: **551**.
- **TAQ intraday data.**
  - Used for spread measures, price impact, intraday volatility, and intraday autocovariance.
  - Period: **1993–2014**.
  - Number of distraction events: **225**.
- **Sample restrictions.**
  - Common stocks only (CRSP share codes 10 or 11).
  - Penny stocks excluded (closing price $< \$1$).
  - TAQ quotes cleaned following Holden and Jacobsen (2014).

2.5 Key observed variables

- **Trading activity.**
  - log($\$volume$),
  - log(trade size),
  - log($\#stocks$),
  - log($\#investors$),
  - log(turnover).
- **Market outcomes.**
  - market return,
  - absolute return,
  - price range,
  - intraday volatility,
  - intraday autocovariance.
- **Liquidity measures.**
  - overall spread measures: closing quoted spread, average quoted spread, effective spread;
  - adverse-selection measures: Amihud illiquidity, price impact, absolute trade imbalance, Kyle's lambda;
  - inventory-cost measure: realized spread.

2.6 Proxies for retail ownership and investor "biasedness"

- **Retail-ownership proxies for stocks.**
  - small market capitalization,
  - low stock price,
  - low institutional ownership.
- **Investor-biasedness proxies.**
  - gender (single male versus single female),
  - online versus offline trading,
  - portfolio concentration,
  - trading intensity,
  - losses / poor performance,
  - Goetzmann–Kumar overconfidence proxy,
  - media susceptibility (buying stocks recently covered in newspapers).
- **Interpretation.** These proxies are used to test whether the investors most affected by distraction are the ones who look most like **noise traders**.

2.7 Timing and event-study setup

- **Seasonality adjustment.** All main time-series variables are first purged of year effects, calendar-month effects, and day-of-week effects (where month and weekday effects may vary by year).
- **Event window.** For each event, the estimation window consists of all **noneconomic trading days** in a **200-day centered window** around the event.
- **Economic meaning.** The paper compares high-news-pressure noneconomic days to nearby noneconomic days, which helps isolate the attention shock from ordinary macroeconomic news.

3 Models and empirical specifications (all equations + interpretation)

3.1 Event-study abnormal outcome

For any variable X, the paper defines the event-day abnormal value as

$$\text{abnormal } X = X_{t=0} - \text{Average}\big(X_{0<|t|<101\,\&\,\text{noneconomic}}\big)\,.$$

Interpretation. The event-day outcome is compared with the average of nearby *noneconomic* trading days. This is the paper's basic quasi-experimental design: compare a distraction day to otherwise similar days that are not contaminated by economic news.

3.2 Theoretical mechanism: distraction as a drop in noise-trading variance

The paper interprets distraction as a temporary fall in the variance of noise trading, that is, a drop in

$$\text{Var(noise trading)}.$$

Using an extension of the Kyle (1985) model with **risk-averse market makers**, the paper derives four qualitative predictions:

$$\Delta\text{Trading Volume} < 0, \qquad \Delta\lambda > 0, \qquad \Delta\text{Volatility} < 0, \qquad \Delta\text{Autocovariance} > 0.$$

Interpretation.

- **Volume falls** because noise traders trade less, and informed speculators also scale back because they have fewer noisy orders behind which to hide.
- **Liquidity worsens** because lower noise-trading variance raises adverse-selection risk for market makers.
- **Volatility falls** because fewer noise trades generate fewer temporary price shocks.
- **Autocovariance rises** (becomes less negative) because there are fewer short-run price reversals.

3.3 Two channels for liquidity

The paper emphasizes that liquidity moves through two opposing forces.

Adverse-selection channel. When noise traders are active, informed traders can hide among them. If noise trading falls, informed orders become easier to infer, so market makers widen spreads and increase price impact.

Inventory-risk channel. If lower noise trading also lowers the future risk of carrying inventory, market makers might need a smaller inventory premium, which would improve liquidity.

Interpretation. The paper's central claim is that for **short-lived distraction shocks**, the **adverse-selection effect dominates**. That is why liquidity deteriorates when distraction reduces noise trading.

3.4 Core market-quality measures

Quoted spread. The quoted bid-ask spread is

$$\text{Spread} = 2\frac{\text{Ask} - \text{Bid}}{\text{Ask} + \text{Bid}}.$$

Interpretation. This is the standard percentage quoted spread around the midpoint. A higher value means worse liquidity.

Effective spread. For a transaction at price P with prevailing midpoint M,

$$\text{Effective Spread} = 2\frac{|P - M|}{M}.$$

Interpretation. This measures the actual ex post trading cost faced by the trader relative to the prevailing midpoint.

Price impact. Let M be the midpoint one second before the trade and M_5 the midpoint five minutes after the trade. Then

$$\text{Price Impact} = 2\frac{M_5 - M}{M_5}.$$

Interpretation. This captures the permanent component of the price change associated with the trade and is intended to isolate the adverse-selection part of illiquidity.

Absolute trade imbalance. If buy and sell dollar volumes during the day are B and S, then

$$\text{Absolute Trade Imbalance} = \frac{|B - S|}{B + S}.$$

Interpretation. More one-sided order flow implies a stronger informational asymmetry between traders and market makers.

Lambda. Kyle's lambda is estimated from the regression

$$r_{5m} = \alpha + \lambda S_{5m} + \varepsilon_{5m},$$

where r_{5m} is the return over a five-minute interval and S_{5m} is signed order flow over that interval.

Interpretation. A larger λ means that order flow moves prices more strongly, so liquidity is worse.

Realized spread. Let $I_{\text{buy/sell}} = 1$ for a buy and -1 for a sell. Then

$$\text{Realized Spread} = 2 I_{\text{buy/sell}} \frac{P - M_5}{M_5}.$$

Interpretation. This is the noninformational part of the effective spread that accrues to liquidity suppliers as compensation for inventory and order-handling costs.

3.5 Trading-activity decomposition

The paper decomposes aggregate trading activity into:

- **extensive margin:** $\log(\#investors)$, whether investors trade at all;
- **search / breadth margin:** $\log(\#stocks)$, how many stocks they trade conditional on trading;
- **intensive margin:** log(trade size), how aggressively they trade conditional on trading.

Interpretation. This decomposition lets the authors ask where attention matters most in the decision process. The strongest retail effect appears on the **extensive margin**, which supports models with a **fixed attention cost** of entering the market.

3.6 Cross-sectional event regressions

To test whether the different market outcomes reflect the same underlying shock, the paper runs cross-sectional regressions across distraction events of the form

$$Y_e = \alpha + \beta \, \text{Abnormal Trading}_e + \varepsilon_e,$$

where Y_e is abnormal volatility, autocovariance, or liquidity for event e.

Interpretation. If distraction really works through reduced noise trading, then events with larger trading drops should also have larger reductions in volatility and larger increases in illiquidity. This is exactly what the paper finds.

3.7 Heterogeneity designs

The paper estimates the same event-study moments after sorting:

- stocks into terciles by **firm size**, **stock price**, and **institutional ownership**;
- investors into groups by **biasedness**;
- small stocks into **tech versus nontech** during the Internet era;
- small stocks into terciles of **algorithmic-trading intensity** in the post-2007 era.

Interpretation. These heterogeneity designs are not secondary robustness checks; they are central to the mechanism. The paper predicts the strongest distraction effects where **noise trading is most important**, and the data match that prediction.

4 Identification and instruments (what makes the estimates credible)

4.1 Core identification idea

There is no traditional instrumental-variable design here. Identification comes from a **quasi-natural experiment**: sensational, high-news-pressure, **noneconomic** news days that exogenously divert investor attention away from the market.

Key idea. If these events are unrelated to stock fundamentals but strongly absorb public attention, then differences between event days and nearby noneconomic days can be interpreted as the causal effect of **noise-trader inattention**.

4.2 Why the events are plausibly exogenous

- The events are selected from TV news rather than stock-market outcomes.
- The paper removes days whose top-news headlines contain economic keywords.
- Macroeconomic releases, business-activity indicators, and newspaper sentiment explain only a small fraction of variation in news pressure.
- The market return itself does not move significantly on distraction days, which is reassuring if the events are truly noneconomic.

Interpretation. The paper is not just finding that "quiet market days are quiet." It is isolating days when attention is pulled away by salient outside events.

4.3 Why the paper can claim the distracted agents are noise traders

- More biased households and institutions reduce trading more on distraction days.
- Small trades fall more than large trades.
- Market effects are strongest in stocks with high retail ownership.

- The data show weak or nonexistent aggregate effects for the whole market but strong effects exactly where noise trading should matter most.

Interpretation. This is the crucial agent-level validation step. The sensational news events do not appear to be distracting everyone equally; they disproportionately affect the traders who look most like **noise traders**.

4.4 Why reverse causality is unlikely

A natural concern is that high news pressure simply occurs on days with *little* economic news, so low trading and volatility could be caused by the absence of economic information rather than distraction.

The paper argues against this on several grounds:

- TV viewership rises sharply on distraction days, which is hard to reconcile with a mere "news drought."
- The estimation window uses only noneconomic days, so event days are compared with other noneconomic days.
- The paper finds no evidence of systematically less firm-specific news on distraction days.
- A placebo using low-news-pressure noneconomic days yields no opposite pattern.

4.5 Alternative explanations the paper rules out

Distracted informed speculators. If informed speculators were the distracted agents, then they would trade less aggressively and **liquidity should improve**, not worsen. That is the opposite of the data.

Distracted market makers. If market makers were distracted, returns should become **more volatile** and **more negatively autocorrelated**. Again, the data go the other way.

Investor sentiment. Many distraction events are emotionally negative. But if sentiment were the main channel, the paper should find significantly negative returns and higher trading. Instead, returns are not significantly negative and trading falls.

Bottom line. The joint pattern—lower volume, lower volatility, less reversal, worse liquidity—matches the theory of **distracted noise traders** much better than the alternatives.

4.6 Why the duration of the shock matters

A particularly strong credibility point in the paper is its emphasis on **transitory** versus **persistent** noise-trading shocks.

- Persistent declines in noise trading mainly reduce future inventory risk and can **improve** liquidity.
- Short-lived declines mainly raise adverse-selection risk *today* and therefore **reduce** liquidity.

Interpretation. This duration argument is what reconciles the paper with prior studies that found the opposite liquidity effect.

5 Tables and figures

5.1 Figure 1: O.J. Simpson verdict as a motivating natural experiment

Read:

- The figure shows intraday NYSE volume on October 3, 1995, when the O.J. Simpson verdict was announced.
- Trading volume collapses immediately after the verdict announcement and then recovers.
- The drop is especially striking in small trades.

Takeaway. This figure is the intuitive starting point of the paper. It shows, almost in real time, that a sensational noneconomic event can pull attention away from the market.

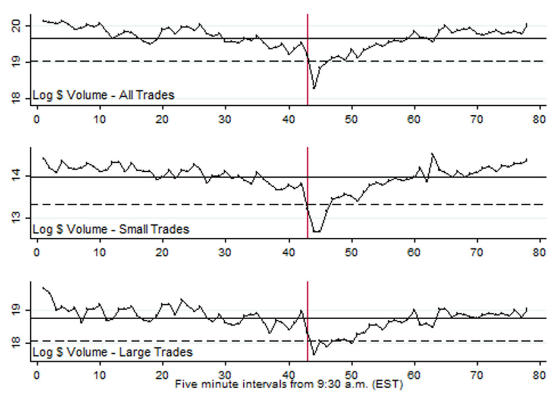

Figure 1. Trading activity during the O.J. Simpson trial verdict. This figure shows the value of aggregate trading volume (in logs) on the New York Stock Exchange on October 3, 1995, the day the verdict of O.J. Simpson's murder trial was announced. The top, middle, and bottom panels display trading volume for, respectively, all, small, and large trades. Trades are sorted into five size groups, where small (large) trades are those in the bottom (top) quintile. The horizontal axis labels five-minute intervals starting from 9:30 a.m. EST. The vertical line marks the announcement time (10 a.m. PST or 1 p.m. EST). The solid horizontal line indicates the average (log) trading volume during that day (excluding the time period from 10:00 to 10:10 a.m.) for the trade size category displayed in the panel. The dashed horizontal line indicates the 5% confidence bound (1.96 times the standard deviation of (log) trading volume during the day). Data for this figure come from Trades and Quotes (TAQ). (Color figure can be viewed at wileyonlinelibrary.com)

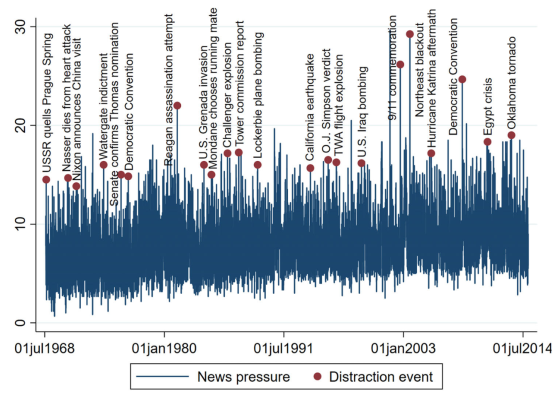

Figure 2. Daily news pressure and distraction events. The blue line shows daily news pressure over the period 1968–2014. The red dots mark a subset of the distraction events that we use in this paper. Specifically, they consist of days when news pressure is the highest in a given year and that have survived our filter for excluding potential economic news (see Section II). (Color figure can be viewed at wileyonlinelibrary.com)

5.2 Figure 2: Time series of news pressure and selected distraction events

Read:

- News pressure fluctuates around a stable mean with occasional large spikes.
- The selected distraction events line up with the major spikes.

Takeaway. The paper's shock variable is not hand-picked around market outcomes. It comes from an externally observed media series.

5.3 Table I: TV viewership and the distraction-event list

Table I

Distraction Events

In Panel A, we report event-study results for TV viewership over the period 1991–2014 (235 distraction events) using data from Nielsen Research. The estimation period includes all trading days without economic news within a 200-day window centered on the distraction event. Column (1) shows the abnormal value of the logarithm of average daily CNN viewership (scaled by the number of U.S. households). Column (2) shows the abnormal value of the logarithm of average 6:30–7:00 p.m. news broadcast viewership for ABC, CBS, and NBC. Below each number, we show the *t*-statistic for the parametric Boehmer, Musumeci, Poulsen's (1991) test in parentheses, and the *z*-statistic for the nonparametric rank test in square brackets. Statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. In Panel B, we provide a partial list of the distraction events used in this paper. For each year, we show the dates of the two distraction events with the highest news pressure (that survive our economic news filter) together with a short description of the accompanying news story.

Panel A: TV Viewership on Distraction Events

(1) CNN viewership (total day)	(2) ABC, CBS, and NBC viewership (6:30–7:00 p.m.)
0.327	0.032
(13.190)***	(6.431)***
[10.729]***	[5.265]***
235	235

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Panel B: Two Highest News Pressure Events (without Economic Keyword in their Headline) per Year

Year	Date	Description	Date	Description
1968	Aug 22	USSR invasion of Czechoslovakia	Nov 1	Vietnam bombing halt
1969	Mar 28	Eisenhower death	Nov 20	Apollo 12 color film from moon
1970	Sep 28	Gamal Abdel Nasser death	Sep 9	Dawson's Field hijackings
1971	Jul 16	Nixon announces China visit	Apr 1	William Calley verdict
1972	Mar 6	Senate questions ITT settlement	May 2	Hoover death
1973	Jan 24	Vietnam ceasefire aftermath	Jul 26	Watergate hearings
1974	Mar 1	Watergate indictments	Feb 13	Solzhenitsyn deportation
1975	Nov 3	Rockefeller decides not to run for VP	May 14	South Vietnam evacuation plans
1976	Jul 13	Democratic Convention	Jun 9	Democratic presidential primaries
1977	Oct 18	West German plane hijacking	Mar 11	Hanafi Siege in Washington, DC
1978	Sep 19	Camp David Accords aftermath	Apr 18	Senate passes Panama Canal treaty
1979	Feb 14	U.S. embassy incident in Tehran	Jan 16	Iranian revolution, Shah flees
1980	Dec 26	Iran hostage crisis	Aug 11	Democratic Convention
1981	Mar 30	Reagan assassination attempt	May 13	Pope assassination attempt
1982	Sep 20	Lebanon massacre	Jun 8	Israel Lebanon invasion
1983	Oct 25	Grenada invasion aftermath	Oct 26	Grenada invasion aftermath
1984	Jul 12	Mondale chooses running mate	Aug 16	John DeLorean verdict
1985	Oct 8	Achille Lauro hijacking	Jun 17	TWA847 hijacking
1986	Jan 28	Challenger explosion	Jan 29	Challenger explosion aftermath
1987	Feb 26	Tower commission report	May 18	USS Stark incident in Persian Gulf
1988	Dec 22	Lockerbie plane bombing	Jul 5	Attorney General Meese resigns
1989	Jan 4	Libyan planes downed	Jul 3	Supreme Court abortion ruling
1990	Aug 8	Address on Iraq's invasion of Kuwait	Aug 16	Persian Gulf crisis talks
1991	Oct 15	Senate confirms Thomas nomination	Jan 10	Preparations for Iraq invasion
1992	May 1	Los Angeles riots	Dec 8	U.S. special forces enter Somalia
1993	Apr 20	Waco sect compound fire	Sep 13	Oslo Accords officially signed
1994	Jan 17	Northridge earthquake	Jan 18	Northridge earthquake aftermath
1995	Oct 3	O. J. Simpson verdict	Apr 20	Oklahoma City bombing
1996	Jul 18	TWA flight explosion	Nov 5	Presidential election aftermath
1997	Sep 5	Princess Diana's funeral	Mar 27	Heaven's Gate sect mass suicide
1998	Dec 16	Iraq missile attack	Dec 18	Clinton impeachment house debate
1999	Mar 25	NATO bombing of Yugoslavia	Apr 23	Littleton school shooting
2000	Nov 22	Presidential election aftermath	Dec 11	Florida recount, legal battles
2001	Oct 12	Anthrax letter attacks	Jun 11	Timothy McVeigh execution
2002	Sep 11	9/11 commemoration	Oct 24	Hurricane Lili
2003	Aug 14	Northeast blackout	Oct 27	California wildfires
2004	Apr 7	Iraq Fallujah uprising	Apr 8	9/11 Commission hearing
2005	Sep 1	Hurricane Katrina aftermath	Jul 7	London bombing
2006	Jan 4	Sago coal mine explosion	Jul 13	Israel Lebanon conflict
2007	Apr 17	Virginia Tech massacre	Aug 2	Minneapolis bridge collapse
2008	Aug 27	Democratic Convention	Nov 3	Presidential election one day before
2009	Dec 28	Northwest Airlines bombing attempt	Jul 7	Michael Jackson memorial service
2010	Jan 15	Haiti earthquake	Mar 22	Health Care reform passed
2011	Jan 31	Egypt crisis	Jan 10	Tucson Arizona shooting
2012	Dec 14	Connecticut school shooting	Jul 20	Aurora movie theatre massacre
2013	May 20	Oklahoma tornado	May 21	Oklahoma tornado aftermath
2014	Jul 17	MH17 plane crash	Jul 18	MH17 plane crash

Read:

- Panel A shows that on distraction days CNN viewership rises by about **33%** and evening-news viewership rises by about **3%**.
- Panel B lists the two highest-news-pressure noneconomic events in each year.

Takeaway. The events are real attention shocks, not merely coding artifacts.

5.4 Table II and Figure 3: Distraction and actual investor trading

Read:

- Retail trading volume falls by about **6.5%**; institutional volume falls by about **4%**.
- For retail investors, the strongest effect is on **the number of investors trading**, down about **5%**.
- Small trades fall by about **2%**; large trades show only a weak and insignificant decline.
- Figure 3 shows that the effect is concentrated on the event day itself, with no clear pre-trend and no catch-up afterward.

Takeaway. Distraction creates a short-lived but sharp fall in trading, and that fall is strongest where retail/noise trading is most prevalent.

5.5 Table III: Which investors are most distracted?

Read:

- More biased investors—single males, online traders, concentrated investors, heavy traders, poor performers, overconfident traders, media-sensitive traders—reduce trading more on distraction days.

Table II
Distraction Events and Retail and Institutional Trading

This table reports event study results for trading activity on distraction days. Panel A reports results for the discount brokerage data over the period 1991–1996 (66 distraction events). Panel B reports results for the institutional trade data over the period 1999–2011 (99 distraction events). Panel C reports results for the ISSM/TAQ transaction data over the period 1991–2000 (105 distraction events). In Panels A and B, log($*volume*) is the logarithm of dollar volume aggregated over all sample investors. Log(*trade size*) is the average across stocks and then across investors of the logarithm of dollar trade size. Log(#*stocks*) is the average across investors of the logarithm of the number of stocks traded. Log(#*investors*) is the logarithm of the number of investors trading. In Panel C, Log($*volume*) is the logarithm of aggregated dollar volume of trade in the respective trade size group. Trades are classified into small trades and large trades based on a procedure described in Hvidkjaer (2006). The estimation period includes all trading days without economic news within a 200-day window centered on the distraction event. In Panels A and B, columns (1) and (2) focus on buys and sells, respectively. Column (3) tests for the difference between buys and sells. Column (4) examines total trades (the sum of buys and sells). In Panel C, columns (1) and (2) show the results for small and large trades, respectively. Column (3) tests for the difference between small and large trades. Below each number, we show the t-statistic for the parametric Boehmer, Musumeci, and Poulsen's (1991) test in parentheses, and the z-statistic for the nonparametric rank test in square brackets. Statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively.

	(1) Buys	(2) Sells	(3) Difference	(4) Total trades
	Panel A: Discount Brokerage Data			
Log($*volume*)	−0.073	−0.056	−0.018	−0.065
	(−2.442)**	(−1.649)	(−0.912)	(−2.221)**
	[−2.635]***	[−1.134]	[−0.476]	[−2.035]**
Log(*trade size*)	−0.017	−0.025	0.008	−0.019
	(−1.593)	(−1.526)	(0.111)	(−1.868)*
	[−1.690]*	[−1.255]	[0.789]	[−1.664]*
Log(#*stocks*)	−0.010	0.003	−0.013	−0.007
	(−2.480)**	(0.841)	(−2.378)**	(−2.625)**
	[−2.622]***	[0.885]	[−2.405]**	[−2.782]***
Log(#*investors*)	−0.0539	−0.0574	0.0034	−0.0508
	(−2.857)***	(−2.061)**	(−0.714)	(−2.609)**
	[−2.693]***	[−2.067]**	[0.214]	[−2.431]**
N	66	66	66	66
	Panel B: Institutional Trade Data			
Log($*volume*)	−0.040	−0.044	0.003	−0.042
	(−2.261)**	(−1.967)*	(−0.266)	(−2.293)**
	[−1.832]*	[−1.871]*	[0.600]	[−2.049]**
Log(*trade size*)	−0.019	−0.012	−0.007	−0.017
	(−2.014)**	(−1.211)	(−0.572)	(−2.066)**
	[−1.892]*	[−0.880]	[−0.848]	[−1.354]
Log(#*stocks*)	−0.014	−0.004	−0.009	−0.014
	(−1.590)	(−0.458)	(−0.858)	(−1.360)
	[−1.759]*	[−0.789]	[−1.044]	[−1.532]

(Continued)

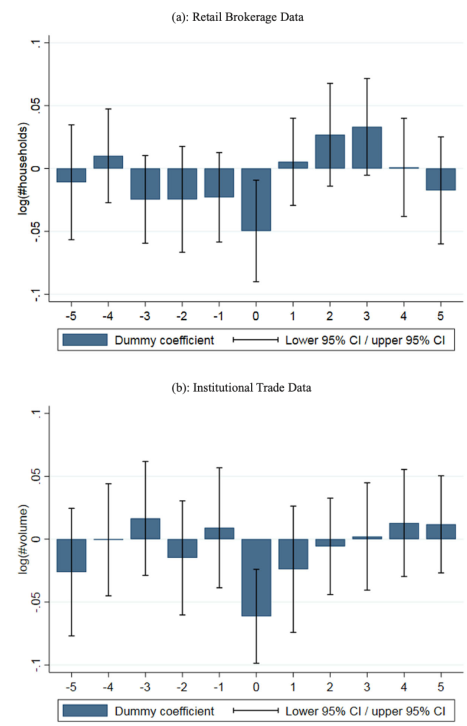

Figure 3. Distraction effect on retail and institutional trading in event time. This graph shows the coefficient estimates and their confidence intervals from regressing measures of (deseasonalized and de-trended) retail and institutional trading on a set of event-time dummies. In Panel A, the dependent variable is the logarithm of number of households trading in the retail brokerage data. In Panel B, the dependent variable is the logarithm of the aggregated trade volume in the institutional trade data. In both panels, the bar at 0 represents the coefficient estimate on a dummy variable indicating distraction days; the bar at −1 is the coefficient estimate on a dummy variable indicating trading days one day prior to distraction days. Confidence bars are based on standard errors adjusted for autocorrelation up to three lags with the Newey-West method. (Color figure can be viewed at wileyonlinelibrary.com)

Table III

Distracting Events and Noise Traders

This table reports event study results for trading activity by different groups of investors. Panel A reports results for the logarithm of the number of households trading in the discount brokerage data over the period 1991–1996 (66 distraction events). Panel B reports results for the logarithm of the aggregated dollar volume in the institutional trade data over the period 1999–2011 (99 distraction events). For each event, the estimation period includes all trading days without economic news within a 200-day window centered on the event date. Each row in Panels A and B represents a different sample split. In row 1, investors are split into single females (column (1)) and single males (column (2)) (only available for Panel A). In row 2, investors are split into those who never traded online (column (1)) and those who did trade online (column (2)) (only available for Panel A). In the other rows, investors are split in halves based on the variable indicated in the row label. Column (3) tests for the difference between above-median investors (column (2)) and below-median investors (column (1)) (or single males and single females for row 1). *PF concentration* is the average concentration (measured by the Herfindahl index) of monthly portfolio holdings (monthly dollar trading volumes across stocks) for retail (institutional) investors. *PF volume* is the retail or institutional investor's average portfolio volume. *PF losses* is the average monthly portfolio return (posttrade return over the subsequent 12 months) times minus one for retail (institutional) investors. *GK-proxy* is the overconfidence proxy proposed by Goetzmann and Kumar (2008) based on the interaction of inverse profits and portfolio turnover (aggregate trading volume) for retail (institutional) investors. *Media susceptibility* is measured as the retail or institutional investor's average propensity to buy stocks covered in the media. Below each number, we show the t-statistic for the parametric Boehmer, Musumeci, and Poulsen's (1991) test in parentheses, and the z-statistic for the nonparametric rank test in square brackets. Statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively.

	(1) Total Trades	(2) Total Trades	(3) Difference
Panel A: Discount Brokerage Data			
(1) Gender	Single female	Single male	Difference
Log(#*investors*)	0.036	−0.093	−0.119
	(0.315)	(−2.790)***	(−2.532)**
	[1.384]	[−1.964]**	[−1.804]*
(2) Online versus offline	Offline	Online	Difference
Log(#*investors*)	−0.039	−0.069	−0.030
	(1.731)*	(−2.063)**	(−1.555)
	[1.838]*	[−2.493]**	[−1.367]
(3) PF concentration	Low	High	Difference
Log(#*investors*)	−0.034	−0.066	−0.032
	(−1.778)*	(−2.671)***	(−0.908)
	[−1.306]	[−2.258]**	[−0.866]
(4) PF volume	Low	High	Difference
Log(#*investors*)	0.001	−0.049	−0.051
	(0.022)	(−2.482)**	(−2.274)**
	[0.361]	[−2.047]**	[−1.313]
(5) PF losses	Low	High	Difference
Log(#*investors*)	−0.036	−0.068	−0.032
	(−1.935)*	(−2.364)**	(−0.880)
	[−1.115]	[−2.380]**	[−1.217]

(Continued)

Table III—*Continued*

	(1) Total Trades	(2) Total Trades	(3) Difference
(6) GK-proxy	Low	High	Difference
Log(#*investors*)	0.001	−0.053	−0.054
	(0.016)	(−2.196)**	(−2.294)**
	[0.533]	[−1.830]*	[−1.319]
(7) Media susceptibility	Low	High	Difference
Log(#*investors*)	−0.028	−0.056	−0.028
	(−1.328)	(−2.389)**	(−0.869)
	[−0.763]	[−2.003]**	[−1.089]
N	66	66	66
Panel B: Institutional Trade Data			
(1) Gender	Single female	Single male	Difference
Log($*volume*)	N/A	N/A	N/A
(2) Online versus offline	Offline	Online	Difference
Log($*volume*)	N/A	N/A	N/A
(3) PF concentration	Low	High	Difference
Log($*volume*)	−0.042	−0.049	−0.007
	(−1.447)	(−2.222)**	(−0.441)
	[−1.358]	[−1.860]*	[−0.552]
(4) PF volume	Low	High	Difference
Log($*volume*)	0.017	−0.043	−0.059
	(0.416)	(−2.309)**	(−2.147)**
	[−0.136]	[−2.119]**	[−1.009]
(5) PF losses	Low	High	Difference
Log($*volume*)	−0.017	−0.056	−0.039
	(−0.923)	(−2.518)**	(−1.345)
	[−0.391]	[−2.325]**	[−1.864]*
(6) GK-proxy	Low	High	Difference
Log($*volume*)	−0.018	−0.044	−0.025
	(−0.431)	(−2.365)**	(−1.487)
	[−1.449]	[−2.223]**	[−0.014]
(7) Media susceptibility	Low	High	Difference
Log($*volume*)	−0.041	−0.062	−0.022
	(−1.686)*	(−2.350)**	(−0.950)
	[−1.282]	[−2.815]***	[−1.036]
N	99	99	99

- This pattern holds for both households and institutions.

Takeaway. The paper does not merely assume that distracted traders are noise traders; it shows that the most distractible traders are exactly those who appear most biased or least skilled.

5.6 Table IV: Market-wide event study

Read:

- At the aggregate market level, the effects are weak.
- There is no significant market-return response.
- Turnover and volume fall slightly, and some illiquidity measures rise, but the broad market response is muted.

Takeaway. The average stock is not where the action is. The mechanism should operate most strongly in stocks with high retail ownership.

5.7 Table V: Sample split by firm size

Read:

- Small stocks show the strongest response: turnover and dollar volume fall significantly.

Table IV

Market Wide Event Study

This table reports (equal-weighted) market wide event-study results for the 551 distraction events in the period 1968–2014. The estimation period includes all trading days without economic news within a 200-day window centered on the event date. All variables are defined in the Appendix. Below each number, we show the t-statistic for the parametric Boehmer, Musumeci, and Poulsen's (1991) test in parentheses, and the z-statistic for the nonparametric rank test in square brackets. Statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively.

(1) Mkt Return	(2) Log(*turnover*)	(3) Log($*volume*)	
−0.017	−0.008	−0.011	
(−0.798)	(−1.124)	(−1.480)	
[−1.175]	[−0.998]	[−1.390]	
551	551	551	
(4) Abs Return	**(5) Price Range**	**(6) Intraday Volatility**	**(7) Intraday Autocovariance**
0.001	−0.007	−0.006	0.005
(1.213)	(0.781)	(−0.226)	(0.269)
[−1.340]	[−0.521]	[−1.144]	[2.105]**
551	551	225	225
(8) Closing Bid-Ask Spread	**(9) Average Bid-Ask Spread**	**(10) Effective Spread**	**(11) Realized Spread**
0.011	0.000	0.009	0.008
(2.258)**	(1.400)	(1.741)*	(1.967)*
[2.051]**	[0.043]	[1.563]	[1.451]
354	225	225	225
(12) Log(*amihud*)	**(13) Price Impact**	**(14) Absolute Trade Imbalance**	**(15) Lambda**
0.009	0.002	0.113	0.135
(2.890)***	(1.002)	(1.637)	(1.943)*
[1.489]	[0.782]	[1.153]	[2.350]**
551	225	225	225

Table V

Sample Split by Firm Size

This table reports event-study results for the 551 distraction events in the period 1968–2014. The estimation period includes all trading days without economic news within a 200-day window centered on the event date. Stocks are sorted into three terciles based on their market capitalization at the end of the last trading day prior to the event. All variables are defined in the Appendix. Columns (1)–(3) show results for terciles 1–3, respectively. Column (4) tests for the difference between tercile 3 and tercile 1. Below each number, we show the t-statistic for the parametric Boehmer, Musumeci, and Poulsen's (1991) test in parentheses, and the z-statistic for the nonparametric rank test in square brackets. Statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively.

	N	(1) Tercile 1	(2) Tercile 2	(3) Tercile 3	(4) Difference
		Panel A: Trading Activity and Volatility			
Trading activity					
Log(*turnover*)	551	−0.023	−0.013	0.003	0.026
		(−3.188)***	(−1.448)	(0.479)	(3.522)***
		[−3.803]***	[−1.724]*	[1.080]	[4.236]***
Log($*volume*)	551	−0.027	−0.015	0.000	0.027
		(−3.601)***	(−1.730)*	(0.092)	(3.652)***
		[−4.125]***	[−2.004]**	[0.673]	[4.254]***
Volatility					
Abs return	551	−0.006	−0.003	0.011	0.017
		(−0.013)	(0.623)	(1.849)*	(2.003)**
		[−1.442]	[−2.146]**	[−0.602]	[1.149]
Price range	551	−0.060	−0.009	0.018	0.077
		(−2.449)**	(0.410)	(2.304)**	(4.526)***
		[−3.505]***	[−1.224]	[1.028]	[4.847]***
Intraday volatility	225	−0.018	−0.009	0.004	0.021
		(−3.322)***	(−0.767)	(1.296)	(4.349)***
		[−3.877]***	[−2.184]**	[0.545]	[3.32]***
Intraday autocovariance	225	0.008	0.005	0.004	−0.004
		(2.240)**	(0.269)	(−0.562)	(−2.418)**
		[2.962]***	[2.133]**	[0.565]	[−1.862]*
		Panel B: Liquidity			
Liquidity—overall					
Closing bid-ask spread	354	0.057	0.011	−0.014	−0.071
		(3.833)***	(1.722)*	(−0.17)	(−2.849)***
		[3.331]***	[1.440]	[0.197]	[−3.265]***
Average bid-ask spread	225	0.042	−0.002	−0.021	−0.063
		(2.607)***	(0.943)	(−0.952)	(−2.229)**
		[1.377]	[−0.500]	[−1.906]*	[−1.891]*
Effective spread	225	0.040	0.014	−0.011	−0.051
		(2.225)**	(2.264)**	(0.747)	(−1.028)
		[1.677]*	[2.154]**	[−0.394]	[−2.014]**
Liquidity—adverse selection					
Log(*amihud*)	551	0.024	0.015	0.000	−0.024
		(3.157)***	(2.913)***	(0.680)	(−1.879)*
		[2.568]**	[1.791]*	[−0.496]	[−2.808]***

(Continued)

Table V—*Continued*

		Panel B: Liquidity			
Price impact	225	0.010	0.003	−0.002	−0.011
		(1.677)*	(1.086)	(0.675)	(−0.626)
		[1.273]	[0.275]	[0.568]	[−1.427]
Absolute trade imbalance	225	0.348	0.213	−0.083	−0.431
		(2.371)**	(1.932)*	(−0.899)	(−2.895)***
		[2.301]**	[1.388]	[−1.965]**	[−2.823]***
Lambda	225	0.818	0.348	−0.077	−0.895
		(2.787)***	(1.903)*	(−0.399)	(−2.084)**
		[3.189]***	[2.365]**	[−0.087]	[−3.308]***
Liquidity—inventory costs					
Realized spread	225	0.035	0.010	−0.009	−0.043
		(2.590)**	(2.019)**	(0.229)	(−1.612)
		[2.119]**	[1.484]	[−0.391]	[−2.172]**

Table VI

Cross-sectional Regressions across Distraction Events

This table reports estimates from cross-sectional regressions of abnormal volatility, return autocovariance, and liquidity measures on abnormal (log) turnover (from CRSP) in the cross-section of distraction events for stocks in the bottom tercile in terms of firm size. For dependent variables from CRSP, we have 551 distraction events in the period 1968–2014. For dependent variables from TAQ, we have 225 distraction events in the period 1993–2014. Abnormal measures are calculated as described in Section II.C, that is, as the realization of the variable on the event date minus its average in the estimation period (trading days without economic news within a 200-day window centered on the event date). All variables are defined in the Appendix. Each cell shows the regression coefficient obtained for regressing the independent variable on the dependent variable. Below each number, we show the t-statistic based on Huber-White standard errors corrected for heteroscedasticity. Statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively.

Dependent Variable	N	Regression coefficient
Volatility		
Abs return	551	0.718***
		(6.24)
Price range	551	2.053***
		(21.15)
Intraday volatility	225	0.278***
		(11.42)
Autocovariance	225	−0.057*
		(−1.76)
Liquidity—overall		
Closing bid-ask spread	354	−0.556***
		(−4.49)
Average bid-ask spread	225	−0.675***
		(−4.90)
Effective spread	225	−44.45***
		(−3.04)
Liquidity—adverse selection		
Log(*amihud*)	551	−0.423***
		(−7.27)
Price impact	225	−0.055
		(−1.28)
Absolute trade imbalance	225	−10.19***
		(−7.74)
Lambda	225	−5.788***
		(−3.05)
Liquidity—inventory costs		
Realized spread	225	−0.462***
		(−2.79)

- Volatility falls and autocovariance rises for small stocks, consistent with fewer transient price-pressure shocks.
- Illiquidity measures rise strongly for small stocks but remain largely unchanged for large stocks.
- In the bottom size tercile, the lambda measure rises especially strongly.

Takeaway. This is the central market-quality table. It shows that the theory works exactly where noise trading should matter most.

5.8 Stock-price and institutional-ownership splits

Read:

- Low-price stocks and low-institutional-ownership stocks show patterns very similar to small stocks.
- The effects generally weaken monotonically across terciles.

Takeaway. The results are not specific to one proxy for retail ownership.

5.9 Table VI: Cross-sectional regressions across events

Read:

- Among small stocks, events with larger abnormal drops in turnover have larger drops in volatility.
- The same events also show larger increases in illiquidity measures.

Takeaway. The paper's several outcome variables are not isolated facts. They move together as manifestations of the same underlying decline in noise trading.

5.10 Table VII: Subperiod analysis

Table VII

Subperiod Analysis

This table reports event-study results carried out for stocks in the bottom tercile in terms of firm size over three subperiods: the "pre-Internet" era spanning 1968–1992 (326 distraction events), the "post-Internet" era spanning 1993–2006 (138 distraction events), and the "algorithmic trading era" spanning 2007–2014 (87 distraction events). The estimation period includes all trading days within a 200-day window centered on the event date. Panel A reports the results for measures of trading activity, return volatility, and return autocovariance; Panel B reports results for liquidity. All variables are defined in the Appendix. Below each number, we show the t-statistic for the parametric Boehmer, Musumeci, and Poulsen's (1991) test in parentheses, and the z-statistic for the nonparametric rank test in square brackets. Statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively.

	(1) 1968–1992	(2) 1993–2006	(3) 2007–2014
Panel A: Trading Activity and Volatility			
Trading activity			
Log(*turnover*)	−0.014	−0.032	−0.043
	(−1.162)	(−2.533)**	(−2.854)***
	[−1.844]*	[−2.792]***	[−2.573]**
	326	138	87
Log($*volume*)	−0.018	−0.038	−0.045
	(−1.544)	(−2.842)***	(−2.678)***
	[−2.052]**	[−3.253]***	[−2.556]**
	326	138	87
Volatility			
Abs return	0.007	0.002	−0.070
	(0.517)	(0.531)	(−1.415)
	[−0.529]	[−0.343]	[−1.883]*
	326	138	87
Price range	−0.034	−0.066	−0.144
	(−1.129)	(−1.704)*	(−1.655)
	[−1.939]*	[−2.19]**	[−2.273]**
	326	138	87
Intraday volatility		−0.012	−0.027
		(−2.113)**	(−2.744)***
		[−2.354]**	[−3.162]***
		138	87
Autocovariance		0.003	0.015
		(0.914)	(2.331)**
		[1.635]	[2.692]***
		138	87
Panel B: Liquidity			
Liquidity —overall			
Closing bid-ask spread	0.074	0.075	0.004
	(2.996)***	(3.389)***	(−0.082)
	[2.387]**	[3.34]***	[−0.927]
	129	138	87

(Continued)

Table VII—*Continued*

Panel B: Liquidity			
Average bid-ask spread		0.086	−0.042
		(3.397)***	(−0.913)
		[3.27]***	[−2.404]**
		138	87
Effective spread		0.060	0.008
		(2.489)**	(0.391)
		[2.388]**	[−0.482]
		138	87
Liquidity—adverse selection			
Log(*amihud*)	0.018	0.036	0.028
	(0.953)	(3.557)***	(1.575)
	[0.768]	[3.455]***	[1.354]
	326	138	87
Price impact		0.013	0.003
		(1.954)*	(−0.21)
		[1.657]*	[−0.487]
		138	87
Absolute trade imbalance		0.514	0.085
		(2.594)**	(0.2)
		[2.551]**	[0.347]
		138	87
Lambda		1.140	0.309
		(2.851)***	(0.789)
		[2.989]***	[1.134]
		138	87
Liquidity—inventory costs			
Realized spread		0.052	0.008
		(2.798)***	(0.539)
		[2.632]***	[−0.174]
		138	87

Read:

- From 1968–1992 to 1993–2006, distraction effects become much stronger for almost all variables.
- From 1993–2006 to 2007–2014, the effects on trading activity, volatility, and price reversals become stronger still.
- In contrast, the liquidity effect largely disappears after 2007.

Takeaway. The impact of noise-trader distraction has not faded over time. It has evolved with changes in technology and market structure.

5.11 Table VIII: Tech versus nontech in the Internet era

Read:

- During 1993–2006, distraction effects are stronger for **tech stocks** than for nontech stocks within the small-stock tercile.
- The differences appear in trading activity, volatility, and liquidity.

Table VIII

Sample Split for Small Stocks by Tech versus Nontech

This table reports event-study results for tech versus nontech stocks (classified based on SIC codes as in Loughran and Ritter (2004)) in the bottom size tercile for the 138 distraction events in the 1993–2006 period, which is marked by the advent of the Internet. The estimation period includes all trading days without economic news within a 200-day window centered on the event date. All variables are defined in the Appendix. Columns (1) and (2) show the results for nontech and tech stocks, respectively. Column (3) tests for the difference between tech and nontech stocks. Below each number, we show the *t*-statistic for the parametric Boehmer, Musumeci, and Poulsen's (1991) test in parentheses, and the *z*-statistic for the nonparametric rank test in square brackets. Statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively.

	N	(1) Nontech Stocks	(2) Tech Stocks	(3) Difference
		Panel A: Trading Activity and Volatility		
Trading activity				
Log(*turnover*)	138	−0.031	−0.043	−0.012
		(−2.614)***	(−2.094)**	(−1.984)**
		[−2.796]***	[−2.108]**	[−1.773]*
Log(*$volume*)	138	−0.033	−0.046	−0.013
		(−2.618)***	(−2.039)**	(−1.940)*
		[−3.014]***	[−2.071]**	[−1.720]*
Volatility				
Abs return	138	0.003	−0.011	−0.014
		(0.440)	(−1.056)	(−1.423)
		[0.358]	[−1.375]	[−1.375]
Price range	138	−0.064	−0.072	−0.008
		(−1.697)*	(−1.408)	(−1.001)
		[−2.109]**	[−1.790]*	[−1.272]
Intraday volatility	138	−0.011	−0.015	−0.004
		(−2.017)**	(−1.797)*	(−1.280)
		[−1.988]**	[−1.730]*	[−1.570]
Intraday autocovariance	138	0.003	0.004	0.001
		(0.811)	(1.392)	(0.582)
		[1.366]	[1.011]	[0.435]
		Panel B: Liquidity		
Liquidity—overall				
Closing bid-ask spread	138	0.073	0.080	0.007
		(3.182)***	(2.030)**	(1.540)
		[3.144]***	[2.274]**	[1.465]
Average bid-ask spread	138	0.085	0.092	0.007
		(3.303)***	(2.644)***	(1.793)*
		[3.231]***	[2.352]**	[1.684]*
Effective spread	138	0.058	0.064	0.006
		(2.384)**	(2.071)**	(1.911)*
		[2.228]**	[2.118]**	[1.634]
Liquidity—adverse selection				
Log(*amihud*)	138	0.034	0.043	0.009
		(3.415)***	(2.532)**	(1.803)*
		[3.312]***	[2.334]**	[0.903]

(Continued)

Table VIII—*Continued*

	N	(1) Nontech Stocks	(2) Tech Stocks	(3) Difference
		Panel B: Liquidity		
Price impact	138	0.010	0.034	0.023
		(1.614)	(1.912)*	(1.173)
		[1.451]	[1.735]*	[1.310]
Absolute trade imbalance	138	0.486	0.629	0.143
		(2.352)**	(2.186)**	(1.599)
		[2.213]**	[1.892]*	[1.455]
Lambda	138	1.002	1.403	0.401
		(2.646)***	(2.241)**	(1.856)*
		[2.698]***	[2.155]**	[1.862]*
Liquidity—inventory costs				
Realized spread	138	0.050	0.059	0.009
		(2.603)***	(2.018)**	(1.779)*
		[2.479]**	[1.920]*	[1.632]

Takeaway. This supports the claim that the Internet amplified distraction effects by emboldening or accelerating the behavior of tech-savvy investors.

5.12 Table IX: Algorithmic trading in the post-2007 period

Table IX

Sample Split for Small Stocks by Quote-to-trade Ratio

This table reports event-study results for the tercile of small stocks for the 87 distraction events in the 2007–2014 period, which is marked by the advent of algorithmic trading. The estimation period includes all trading days without economic news within a 200-day window centered on the event date. Small stocks are sorted into three terciles based on the average quote-to-trade ratio calculated from TAQ data (see Section VI.C for details). All variables are defined in the Appendix. Columns (1)–(3) show results for terciles 1–3, respectively. Column (4) tests for the difference between tercile 3 and tercile 1. Below each number, we show the t-statistic for the parametric Boehmer, Musumeci, and Poulsen's (1991) test in parentheses, and the z-statistic for the nonparametric rank test in square brackets. Statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively.

		Quote-to-Trade Ratio			
	N	(1) Tercile 1	(2) Tercile 2	(3) Tercile 3	(4) Difference
Panel A: Trading Activity and Volatility					
Trading activity					
Log(*turnover*)	87	−0.038	−0.045	−0.067	−0.028
		(−2.179)**	(−2.364)**	(−3.149)***	(−2.373)**
		[−2.013]**	[−2.175]**	[−3.047]***	[−2.383]**
Log($*volume*)	87	−0.045	−0.044	−0.067	−0.022
		(−2.271)**	(−2.061)**	(−4.448)***	(−2.071)**
		[−2.111]**	[−1.972]**	[−3.843]***	[−2.175]**
Volatility					
Abs return	87	−0.063	−0.095	−0.094	−0.031
		(−0.925)	(−1.931)*	(−2.373)**	(−1.892)*
		[−1.503]	[−1.795]*	[−2.383]**	[−1.375]
Price range	87	−0.102	−0.179	−0.215	−0.113
		(−0.851)	(−2.117)**	(−3.029)***	(−2.450)**
		[−1.710]*	[−2.239]**	[−2.810]***	[−2.569]**
Intraday volatility	87	−0.020	−0.029	−0.038	−0.018
		(−1.631)	(−2.787)***	(−4.308)***	(−2.316)**
		[−2.341]**	[−2.620]***	[−3.784]***	[−2.417]**
Intraday autocovariance	87	0.014	0.012	0.017	0.003
		(1.790)*	(1.225)	(1.861)*	(0.451)
		[2.154]**	[1.443]	[2.434]**	[0.588]
Panel B: Liquidity					
Liquidity—overall					
Closing bid-ask spread	87	0.026	−0.012	0.003	−0.023
		(0.212)	(−0.862)	(−0.180)	(−0.366)
		[−0.703]	[−1.210]	[−0.491]	[−0.063]
Average bid-ask spread	87	0.036	−0.053	−0.050	−0.086
		(0.235)	(−0.800)	(−1.306)	(1.598)
		[0.263]	[−2.446]**	[−2.157]**	[1.760]*
Effective spread	87	0.034	0.005	0.004	−0.030
		(1.728)*	(−0.152)	(0.139)	(−1.581)
		[0.914]	[−0.690]	[−0.584]	[−1.524]

(Continued)

Table IX—*Continued*

Panel B: Liquidity					
Liquidity–adverse selection					
Log(*amihud*)	87	0.040	0.035	0.032	−0.008
		(2.337)**	(1.533)	(1.911)*	(−0.670)
		[1.714]*	[1.579]	[1.811]*	[−0.504]
Price impact	87	0.013	−0.005	0.008	−0.005
		(0.219)	(−0.726)	(0.816)	(−0.571)
		[0.969]	[−1.253]	[0.720]	[−0.313]
Absolute trade imbalance	87	0.427	0.123	0.048	−0.379
		(1.533)	(0.248)	(0.158)	(−1.098)
		[1.464]	[0.042]	[0.106]	[−1.215]
Lambda	87	0.247	0.211	0.592	0.344
		(0.832)	(0.218)	(0.864)	(−0.071)
		[0.487]	[0.178]	[1.117]	[0.309]
Liquidity—inventory costs					
Realized spread	87	0.027	0.013	−0.010	−0.036
		(1.721)*	(0.146)	(−0.356)	(−1.711)*
		[1.257]	[−0.415]	[−0.961]	[−2.044]**

Read:

- In small stocks with more algorithmic trading, the post-2007 distraction effect on trading activity and volatility is stronger.
- But the liquidity effect is weaker.

Takeaway. The paper interprets this as evidence that algorithms and high-frequency traders interact with noise traders in ways that amplify some outcomes (volume, volatility, reversals) while muting others (liquidity).

6 Short summary

- **Method.** Use sensational noneconomic TV-news shocks to distract investors and run event studies around those days.
- **Identification.** Compare high-news-pressure noneconomic days to nearby noneconomic days, then show that the traders most affected are precisely the most noise-trader-like.

- **Application.** U.S. equity-market trading, liquidity, and volatility from 1968 to 2014, combining brokerage data, institutional transaction data, CRSP, and TAQ.
- **Main result.** When noise traders are distracted, they trade less; in high-retail-ownership stocks, liquidity worsens, volatility falls, and price reversals diminish. These effects strengthen after the Internet and interact importantly with algorithmic trading after 2007.

7 Conclusion

7.1 Core conclusion

The paper shows that **noise-trader attention is an important determinant of market quality**. When noise traders are distracted by sensational news, markets become quieter but also less liquid, especially in stocks where retail ownership is high.

7.2 Economic intuition

Noise traders create cover for informed traders. Their trades make order flow noisier, which lowers market makers' adverse-selection problem. Once those noise trades disappear, the remaining order flow becomes more informative, so market makers protect themselves by demanding wider spreads or stronger price concessions. At the same time, fewer noise trades mean fewer temporary price pressures, so volatility and reversals decline.

7.3 What this paper adds

The paper's main contribution is to show that the effect of noise trading on liquidity depends on the **persistence of the shock**. A permanent decline in noise trading can improve liquidity through lower inventory risk, but a short-lived decline worsens liquidity through stronger adverse selection. That insight helps reconcile different findings in the market-microstructure literature.

7.4 Takeaway

This is a paper about more than attention. It is really about the economic role of noise traders. The central lesson is that noise trading is not merely wasteful activity. It is part of what makes markets liquid. When noise traders stop trading, even briefly, prices become less noisy but trading becomes harder.